

1. Administrative & Study Identification

Full Study Title A Phase 1/2 Open-label, Multicenter, Dose Escalation and Dose Expansion Study of the Safety, Tolerability, and Pharmacokinetics of HPN328 Monotherapy and HPN328 With Atezolizumab or Ifinatamab Deruxtecan (I-DXd) in Patients With Advanced Cancers Associated With Expression of Delta-like Canonical Notch Ligand 3 (DLL3) **Short Title/Acronym** MK-6070-001 / HPN328-4001 **Protocol Number** MK-6070-001 (Other ID: HPN328-4001) **NCT Number** NCT04471727 **Sponsor Name** Harpoon Therapeutics, Inc., a subsidiary of Merck & Co., Inc. (Rahway, New Jersey USA) **Investigational Product** Gocatumig (MK-6070 / HPN328), Atezolizumab (Tecentriq), and Ifinatamab Deruxtecan (I-DXd) **Phase of Development** Phase 1 / Phase 2 **Medical Monitor** Harpoon Therapeutics Clinical Operations (hpn328_4001ctgov@harpoontx.com)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the maximum tolerated dose (MTD), recommended dose for expansion (RDE), safety, tolerability, and pharmacokinetics (PK) of gocatumig (HPN328/MK-6070) alone, and in combination with atezolizumab or I-DXd in participants with advanced cancers. **Secondary Objectives:** To evaluate preliminary efficacy, including Objective Response Rate (ORR), and immunogenicity of the study drug combinations in the target patient populations.

2.2 Background & Rationale Small-cell lung cancer (SCLC) and other neuroendocrine carcinomas (NEC) are aggressive malignancies frequently associated with the expression of Delta-like Canonical Notch Ligand 3 (DLL3). Gocatumig (HPN328/MK-6070) is a DLL3-targeting, tri-specific, half-life extended T-cell engager designed to redirect T-cells to attack DLL3-expressing tumor cells. Combining MK-6070 with immune checkpoint inhibitors (e.g., atezolizumab) or antibody-drug conjugates (e.g., I-DXd) may offer improved clinical benefit and anti-tumor activity for patients with relapsed/refractory advanced cancers who have progressed on standard therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** Recruiting **Control Type:** Single-arm (Dose Escalation and Expansion) **Blinding:** Open-label **Randomization:** N/A (Sequential dose-escalation and cohort-based expansion)

3.2 Planned Enrollment & Duration Target \$\$\$: 232 participants (Estimated) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 14-Dec-2020 **Estimated Primary Completion:** 28-Jan-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed malignancy associated with expression of Delta-like Canonical Notch Ligand 3 (DLL3). 2. Small cell lung cancer (SCLC) which is relapsed/refractory following at least 1 prior line of systemic therapy that included platinum-based chemotherapy. 3. Neuroendocrine Prostate Cancer (NEPC; de novo or treatment-emergent) which is relapsed/refractory to standard systemic therapy. 4. High-grade neuroendocrine tumor types other than SCLC and NEPC, with at least one of the following: disease that is relapsed/refractory to standard systemic therapy, disease for which standard therapy does not exist, or disease for which standard therapy is not considered appropriate by the Investigator. 5. Must be able to provide an archival tissue sample or fresh biopsy tissue sample.

4.2 Exclusion Criteria 1. Untreated central nervous system (CNS) metastases, glioma or other primary CNS malignancy, spinal cord compression, symptomatic/uncontrolled epidural disease, or a history of intracranial or spinal cord hemorrhage. 2. Active neurologic paraneoplastic syndrome. 3. Uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures (e.g., biweekly or more frequently). 4. Active or history of autoimmune disease or immune deficiency (including myasthenia gravis, myositis, autoimmune hepatitis, systemic lupus erythematosus, rheumatoid arthritis, psoriatic arthritis, inflammatory bowel disease, antiphospholipid antibody syndrome, Wegener granulomatosis, Sjögren's syndrome, Guillain-Barré syndrome, or multiple sclerosis). 5. Ongoing treatment with systemic immunosuppressive medications (e.g., prednisone >10mg/day or equivalent) within 2 weeks prior to initiation, or systemic immunostimulatory agents (e.g., interferon, IL-2) within 4 weeks. 6. Active viral hepatitis (HAV IgM positive, HBV HBsAg positive, or HCV antibody positive) or uncontrolled HIV infection. Patients with treated/undetectable viral loads or well-controlled HIV are eligible. 7. History of clinically significant cardiovascular disease (e.g., myocardial infarction, symptomatic congestive heart failure NYHA > class II, or uncontrolled cardiac arrhythmia) or arterial thrombosis within 6 months of the first dose. 8. History of allogeneic stem cell transplant or solid-organ transplant. 9. History of severe anaphylactic reactions to chimeric or humanized antibodies or fusion proteins. 10. History of interstitial lung disease, idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or active pneumonitis on screening CT. 11. Treatment with other investigational drugs within 3 weeks of scheduled dosing (or 5 half-lives of the drug, whichever is shorter).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Gocatamig (HPN328) Monotherapy:** Step dosing (priming dose) in the first cycle to mitigate cytokine release syndrome (CRS), followed by target doses administered once weekly (QW) or every 2 weeks (Q2W). * **Combination with Atezolizumab:** Gocatamig Q2W + Atezolizumab Q4W. * **Combination with I-DXd:** Gocatamig Q2W + I-DXd Q3W (on Day 1 and Day 22 of each 42-day cycle).

Route of Administration: Intravenous (IV) Infusion. **Duration of Treatment:** Up to approximately 4 years, or until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care. **Prohibited:** Systemic immunosuppressive medications (except protocol-required pre-medications), systemic immunostimulatory agents (e.g., interferon, IL-2), and concurrent investigational agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days -28 to 0	Informed Consent, Medical History, DLL3 Expression Confirmation, Imaging, Safety Labs
Baseline	Cycle 1, Day 1	First Dose (Priming Dose for IV Gocatamig), PK/PD Sampling, Vital Signs
Treatment/Interim	QW, Q2W, or Q3W	IV Infusions, Toxicity Monitoring (specifically CRS/ICANS), Tumor Imaging
End of Study	Up to ~4 Years	Final Assessment, Safety Follow-up, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Percentage of participants who experience an adverse event (AE). 2. Percentage of participants who discontinue due to an adverse event (AE). 3. Number of participants with a dose-limiting toxicity (DLT) following treatment with HPN328 as monotherapy or in combination with atezolizumab or I-DXd. **Measurement Method:** NCI CTCAE version 5.0 (incorporating ASTCT grading criteria specifically for CRS and ICANS).

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) based on RECIST v1.1 (Prostate Cancer Clinical Trials Working Group 3 [PCWG3] for participants with NEPC). * Extra-cranial Objective Response Rate (EC-ORR) based on modified RECIST v1.1. * Best Overall Response (BOR). * Immunogenicity and anti-drug antibodies.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring utilizing NCI CTCAE v5.0 criteria. Special monitoring procedures are implemented for Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS), graded per ASTCT criteria. **Serious Adverse Events (SAEs):** Routine expedited reporting protocols apply. Dose escalation cohorts utilize a safety run-in period to assess DLTs prior to cohort expansion. **Data Monitoring Committee (DMC):** Internal safety review committees evaluate DLTs prior to any dose escalations.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Population (all participants receiving ≥ 1 dose) and Efficacy Evaluable Population. **Sample Size Justification:** Estimated $N=232$ participants distributed across dose escalation and expansion cohorts. Dose escalation in part 1 utilizes Bayesian optimal interval dosing principles. **Handling of Missing Data:** Standard methodologies for Phase 1/2 oncology trials (e.g., censoring for time-to-event endpoints).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard sponsor-directed site monitoring to ensure protocol adherence and safety tracking. **Audit Trail:** Electronic Data Capture (EDC) systems utilized for eCRF locking and compliance tracking.

11. Study Identifiers & Governance

Primary ID: MK-6070-001 (HPN328-4001) **Registry Number:** NCT04471727 **Sponsor/Lead Organization:** Harpoon Therapeutics, Inc., a subsidiary of Merck & Co., Inc. (Rahway, New Jersey USA) **Study Title:** A Study in Participants With Advanced Cancers Associated With Expression of DLL3 **Official Regulatory Title:** A Phase 1/2 Open-label, Multicenter, Dose Escalation and Dose Expansion Study of the Safety, Tolerability, and Pharmacokinetics of HPN328 Monotherapy and HPN328 With Atezolizumab or Ifinatumab Deruxtecan (I-DXd) in Patients With Advanced Cancers Associated With Expression of Delta-like Canonical Notch Ligand 3 (DLL3)

12. Clinical Framework

Primary Condition: Small-Cell Lung Cancer (SCLC) and Neuroendocrine Carcinoma (NEC) **Diagnosis Threshold:** Relapsed/

refractory, metastatic status **Medical Methodology:** T-cell engager targeted therapy (DLL3) +/- immune checkpoint inhibitor / antibody-drug conjugate **Optimization Target:** Disease control and tumor regression (ORR)

13. Study Procedures & Methodology

Management Pathway: Step-dosing (priming dose) protocols to mitigate CRS risk. **Education Protocol (HCP):** Site initiation visits and specialized training on ASTCT criteria for grading/managing CRS and ICANS. **Education Protocol (Patient):** Standard informed consent and symptom reporting (e.g., fever, neurological changes). **Quality Audit Mechanism:** Independent safety reviews between dose escalations.

14. Observation & Follow-Up Schedule

Total Duration: Up to ~4 years **Onsite Visit Cadence:** Weekly or Bi-weekly during treatment cycles for IV infusion. **Remote Monitoring Frequency:** Between treatment cycles, as clinically indicated. **Checklist Review Interval:** Regular tumor assessments per RECIST v1.1 (typically every 6-8 weeks).

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				